

Sinhala ASR Text Cleaner

batch_cleaner_logs.py — User Guide

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1. Overview

batch_cleaner_logs.py is an automated, directory-based batch processing script that cleans noisy Sinhala ASR output from SLT Mobitel call center recordings.

Unlike previous versions, this script acts as a headless microservice with a self-setup system. On first run it automatically installs all dependencies and creates a .env file for the API key. It then scans a designated input folder for .txt files, groups them into configurable batches, sends each batch in a single Gemini API call to reduce cost, saves cleaned texts to an output folder, and archives the originals to prevent duplicate processing. A permanent audit trail (process.log) is generated for all operations.

1.1 What it Does

- Fixes corrupted and garbled Sinhala Unicode characters
- Merges broken or split words back into correct form
- Removes random noise characters from ASR output
- Preserves English words (bill, payment, connection, SMS, etc.)
- Keeps numbers exactly as they appear
- Automatically installs python-dotenv and google-genai on first run
- Auto-creates .env file and guides the user to fill in the API key
- Automatically retries if the Gemini server is temporarily busy
- Fixes garbled Sinhala display on Windows terminals automatically
- Batch Grouping: Sends multiple .txt files per API call to reduce prompt token overhead and lower cost
- Archive Pattern: Moves successfully processed raw files into a separate archive folder to prevent reprocessing
- Smart Renaming: Automatically appends _cleaned to the output files (e.g., call1_cleaned.txt)
- Fault Tolerance: If one file fails, the script logs the error and continues processing remaining files
- Audit Logging: Generates a real-time process.log file tracking success, failure, and processing time

1.2 Requirements

Requirement	Version	Notes
Python	3.10 or higher	Download from python.org
python-dotenv	Any	Auto-installed on first run
google-genai	Any	Auto-installed on first run
Gemini API Key	Paid tier	From aistudio.google.com/apikey
Internet	Required	For Gemini API calls

**TIP**

The script automatically installs all required packages (python-dotenv, google-genai) if they are not already installed. No manual pip install needed.

2. Setup — First Run (Auto Setup)

This version uses a fully automated setup. You do not need to paste an API key directly into the code. On first run, the script guides you through everything.

2.1 Setting the API Key

Run the script for the first time:

```
python batch_cleaner_logs.py
```

The script will automatically create a .env file in the same folder and stop with instructions:

```
=====
[✓] .env file created successfully!
=====

📁 Location: ...\\Text_Cleaner_Batch_Processing\\.env

👉 Open the .env file and replace:
  GEMINI_API_KEY=YOUR_API_KEY_HERE
  with your real Gemini API key.

Then run this script again. [✓]
=====
```

Open the .env file and edit it:

```
# Before:
GEMINI_API_KEY=YOUR_API_KEY_HERE

# After:
GEMINI_API_KEY=AIzaSy..... <- your real key
```



IMPORTANT

Never share your API key. The .env file is listed in .gitignore and will never be pushed to GitHub. Do not paste your key directly into batch_cleaner_logs.py.

2.2 Selecting the Model (Optional)

Open batch_cleaner_logs.py and locate the SELECT_MODEL variable:

```
SELECT_MODEL = "gemini-2.5-flash" # Default model
```

Model Name	Input \$/1M	Output \$/1M	Best For
gemini-2.5-flash-lite	\$0.10	\$0.40	Cheapest, normal quality
gemini-2.5-flash	\$0.30	\$2.50	Better quality, slightly higher cost
gemini-3-flash-preview	\$0.50	\$3.00	Best quality for difficult, noisy text

Model Name	Input \$/1M	Output \$/1M	Best For
gemini-2.5-pro	\$1.25	\$10.00	Overkill for this task, not recommended

2.3 Setting Batch File Count

This is the key new setting for cost reduction. Locate the `Number_of_TXT_Files_to_Send` variable just below `SELECT_MODEL`:

```
# How many .txt files to send together in one API call
# Saves cost by reusing the prompt once instead of repeating it per file.
# Recommended: 3 to 5. Max: 10 (higher risks Gemini mixing outputs).
Number_of_TXT_Files_to_Send = 3
```

Example with 9 files and batch size = 3:

```
API Call 1 --> [FILE_1, FILE_2, FILE_3] (1 prompt sent)
API Call 2 --> [FILE_4, FILE_5, FILE_6] (1 prompt sent)
API Call 3 --> [FILE_7, FILE_8, FILE_9] (1 prompt sent)

Result: 3 API calls instead of 9 -- saves 6 prompt repetitions
```



A batch size of 3 is the recommended starting point. It balances cost savings with reliability. Increasing to 5 saves slightly more but increases the chance that Gemini mixes up outputs between files.

2.4 Directory Initialization (Automatic)

You do not need to manually create folders. The script auto-creates three folders next to itself on first run:

- `input_audios/` — Drop your noisy ASR .txt files here
- `cleaned_outputs/` — Cleaned files are saved here with `_cleaned` suffix
- `processed_archive/` — Original raw files are moved here after processing

3. How to Run & Process Files

Step 1: Place your noisy ASR transcripts (.txt files) into the input_audios/ folder.

Step 2: Open Command Prompt or Terminal and navigate to the script folder:

```
cd path\to\Text_Cleaner_Batch_Processing
```

Step 3: Run the script:

```
python batch_cleaner_logs.py
```



NOTE

On some systems use python3 instead of python if the command is not found.

Step 4: Check the output. Once complete:

- Open cleaned_outputs/ to retrieve your corrected files
- Open processed_archive/ to find your original raw files securely stored
- Open process.log to see the full audit trail (success, failure, timing)

4. Cost Estimation

4.1 Measured Token Counts

These numbers were measured on a real 2-minute SLT call transcript:

Item	Tokens	Notes
Input (prompt + noisy text)	805 tokens	Original measured value
Input (after prompt optimization)	656 tokens	Shortened prompt, same quality
Output (cleaned text)	~550 tokens	Estimated, similar length to input

4.2 Previous Approach — One File per API Call

Based on 656 input tokens and 550 output tokens per file. Each file = one separate API call.

Model	Input \$/1M	Output \$/1M	Total/Month (USD)	Monthly Cost (LKR ~300)
gemini-2.5-flash-lite	\$0.10	\$0.40	\$85.68	~Rs. 25,704
gemini-2.5-flash	\$0.30	\$2.50	\$471.54	~Rs. 141,462
gemini-3-flash-preview	\$0.50	\$3.00	\$593.40	~Rs. 178,020
gemini-2.5-pro	\$1.25	\$10.00	\$1,896.00	~Rs. 568,800

4.3 New Approach — Multi-File Batching (3 files per call)

With batch size = 3, the prompt (~300 tokens) is sent once per 3 files instead of 3 times. This reduces effective input tokens per file from 656 to 456 tokens (prompt overhead shared across 3 files).

Based on 456 input tokens and 550 output tokens per file. 10,000 files/day, 30 days.

Model	Input \$/1M	Output \$/1M	Total/Month (USD)	Monthly Cost (LKR ~300)
gemini-2.5-flash-lite	\$0.10	\$0.40	\$79.68	~Rs. 23,904
gemini-2.5-flash	\$0.30	\$2.50	\$453.54	~Rs. 136,062
gemini-3-flash-preview	\$0.50	\$3.00	\$563.40	~Rs. 169,020
gemini-2.5-pro	\$1.25	\$10.00	\$1821.00	~Rs. 546,300

**TIP**

gemini-2.5-flash-lite with batch size 3 costs ~Rs. 18,900/month — the most cost-effective option.

4.4 Cost Comparison & Savings

Direct comparison between the previous one-file-per-call approach and the new batched approach (batch size = 3). Based on 10,000 files/day for 30 days.

Model	Old Monthly (USD)	New Monthly (USD)	Saving (USD)	Saving %
gemini-2.5-flash-lite	\$85.68	\$79.68	\$6.00	7.0%
gemini-2.5-flash	\$471.54	\$453.54	\$18.00	3.8%
gemini-3-flash-preview	\$593.40	\$563.40	\$30.00	5.1%
gemini-2.5-pro	\$1896.00	\$1821.00	\$75.00	4.0%

Per-file cost comparison:

Model	Old Cost/File (USD)	New Cost/File (USD)	Old Cost/File (LKR)	New Cost/File (LKR)
gemini-2.5-flash-lite	\$0.000286	\$0.000266	~Rs. 0.09	~Rs. 0.08
gemini-2.5-flash	\$0.001572	\$0.001512	~Rs. 0.47	~Rs. 0.45
gemini-3-flash-preview	\$0.001978	\$0.001878	~Rs. 0.59	~Rs. 0.56
gemini-2.5-pro	\$0.006320	\$0.006070	~Rs. 1.90	~Rs. 1.82

5. Troubleshooting

Error / Problem	Cause	Solution
API key not valid	Wrong or placeholder key	Open .env file and paste your real Gemini API key
503 UNAVAILABLE	Gemini servers temporarily busy	Script auto-retries 3 times. If still fails, wait 5 min and try again
429 RESOURCE_EXHAUSTED	Free tier daily limit hit	Upgrade to paid Gemini API tier at aistudio.google.com
ModuleNotFoundError: genai	google-genai failed to install	Run: pip install google-genai manually
Sinhala shows as ??????	Terminal encoding issue	Script fixes this automatically on Windows. Try running again
python not found	Python not in PATH	Use py batch_cleaner_logs.py on Windows, or python3 on Linux/Mac
Empty output for a file	Gemini returned blank for that label	Try reducing Number_of_TXT_Files_to_Send to 1 for that file
No .txt files found	input_audios/ folder is empty	Place your noisy ASR .txt files into input_audios/ and run again
404 NOT_FOUND	Model name not recognized by API	Check SELECT_MODEL spelling exactly matches the API registry
Script skips a file	Input .txt file is completely empty	Script logs a warning, moves to archive, and continues automatically
.env not found after clone	.env is gitignored (correct)	Run the script once — it will auto-create .env with instructions

6. Quick Reference Card

Task	What to Do
Set API key	Run script once, then edit the auto-created .env file
Change model	Edit SELECT_MODEL = "model-name" in batch_cleaner_logs.py
Set batch size	Edit Number_of_TXT_Files_to_Send = 3 (recommended)
Process files	Drop raw .txt files into input_audios/ folder
Run the script	python batch_cleaner_logs.py
Get an API key	aistudio.google.com/apikey
Get results	Collect files from cleaned_outputs/ folder
Best quality model	gemini-3-flash-preview
Most cost-effective	gemini-2.5-flash-lite + batch size 3
Check errors	Open process.log in any text editor

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